

Network analysis of character relations in the Harry Potter books

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Research Question

Do fictional social networks in long-form narratives exhibit universal topological properties found in real-world social systems, or do narrative constraints produce distinct structural signatures?

Can the network be explained by any preexisting model?

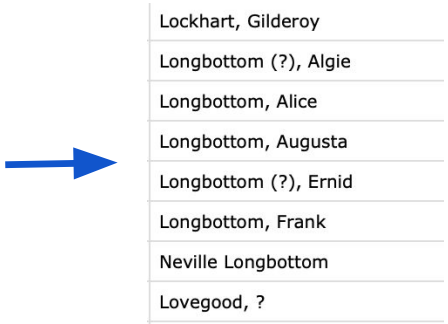
Motivation

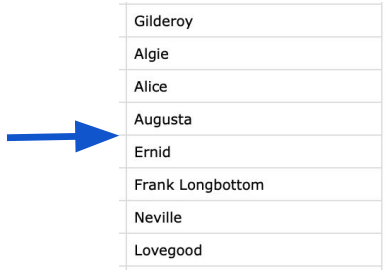
- Fictional social network
- Real world application on storytelling domain
- Similarity with real social networks
- Easily accessible data

Data preprocessing









CSV

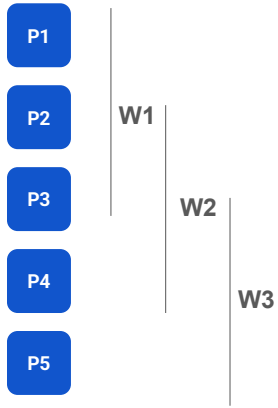
Data description

Nodes: unique character identifiers

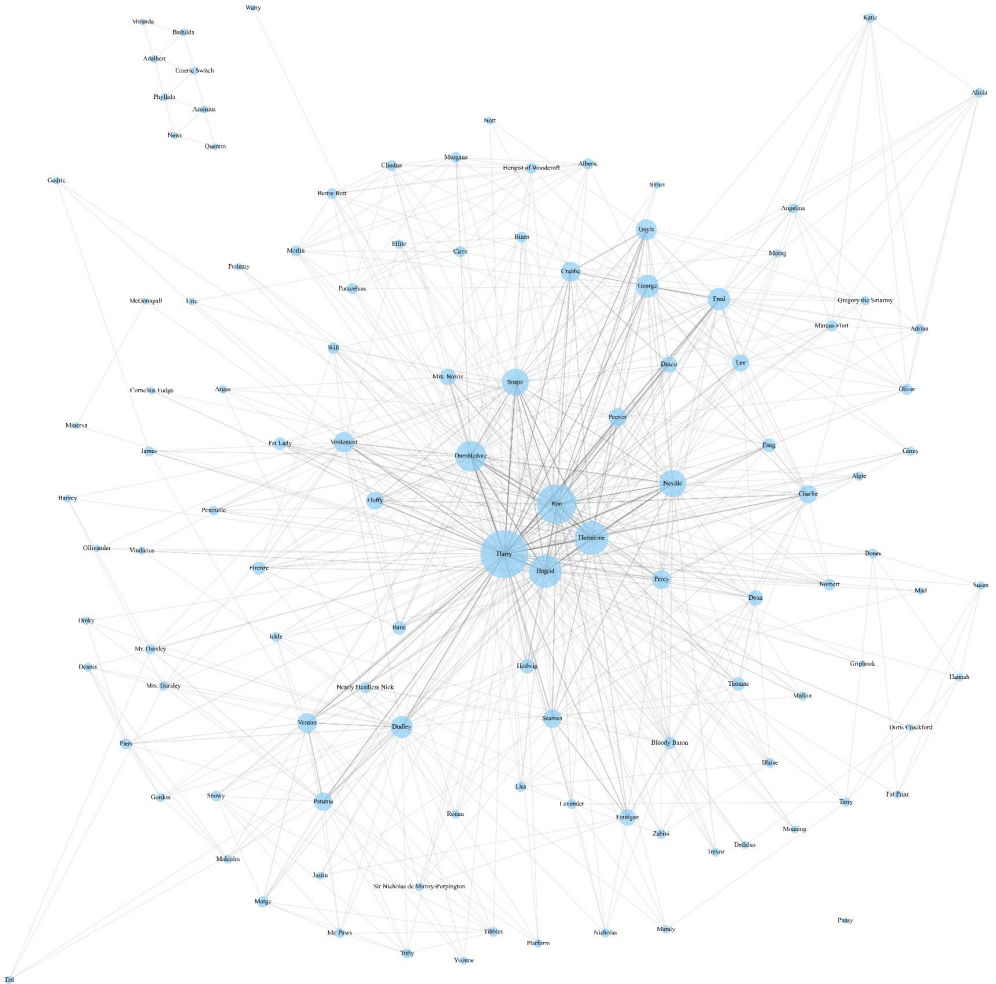
- Size corresponds to degree

Edges: weighted links between characters

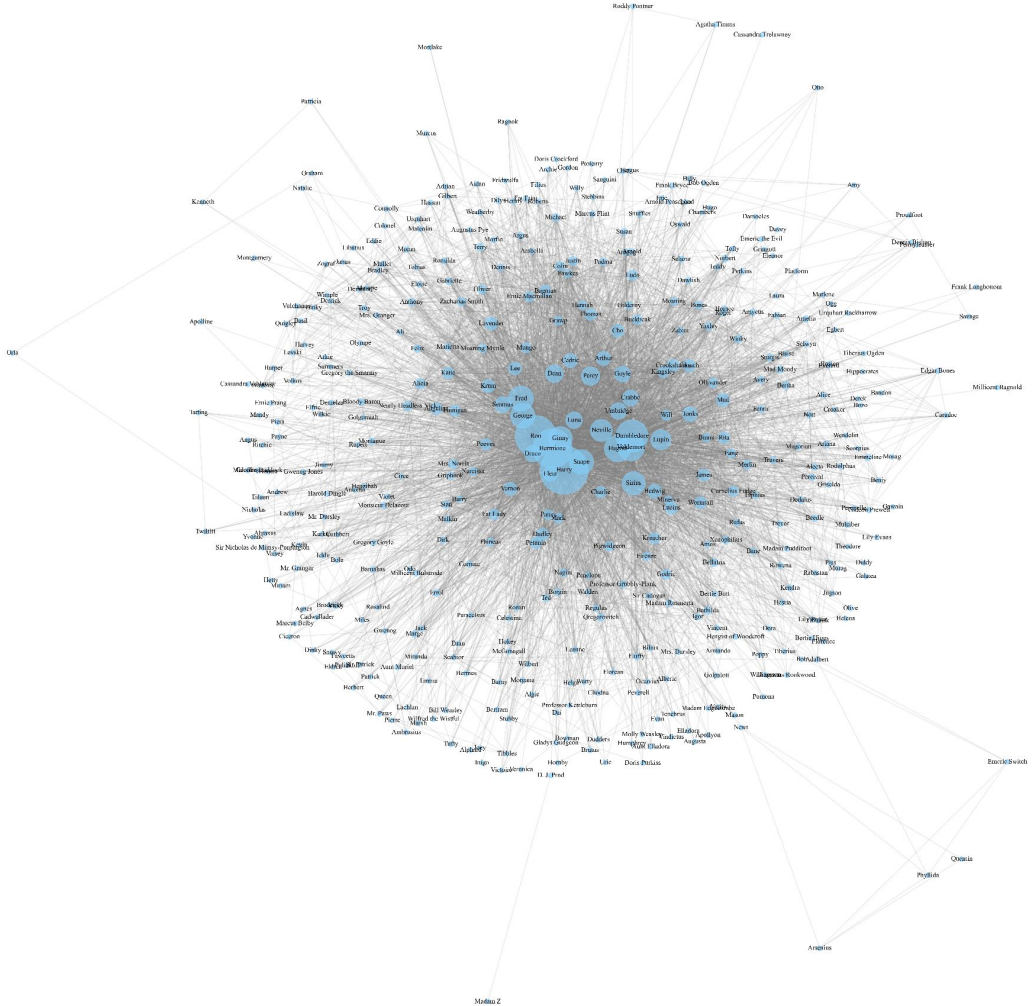
- Total appearances of two characters in three paragraph sliding window



BOOK 01


Harry
Ron
Hermione
Hagrid
Dumbledore
Voldemort
Snape
Neville
Ginny
Fred
George
Percy
Draco
Lucius
Bellatrix
Sirius
Remus
Tonks
Molly
Arthur
Dudley
Vernon
Petunia
Cho
Luna
Dobby
Kreacher
Griphook
Ollivander
Mr. Dursley
Mrs. Dursley
Nearly Headless Nick
Moaning Myrtle
Hedwig
Scabbers
Bloody Baron
Dean
Seamus
Lavender
Parvati
Padma
Cormac McLaggen
Marcus Flint
Gregory the Smarmy
Crabbe
Goyle
Blaise
Millicent
Katie
Alicia
Angelina
Adrian
Oliver
Charlie
Percy
Norbert
Bill
Fleur
Olympe
Madame Maxime
Viktor
Cedric
Diggory
Cho
Fat Lady
Peeves
Filch
Mrs. Norris
Minerva
Quirrell
Quirinus
Firenze
Griphook
Ragnuk
Flitwick
Sprout
Pomona
Slughorn
Horace
Merope
Tom Riddle
Hepzibah Smith
Helga
Rowena
Godric
Salazar
Nicolas
Perenelle
Nicholas
Scrimgeour
Kingsley
Tonks
Lupin
Mundungus
Dedalus
Zabini
Trevor
Pansy
Marcus Belby
Amelia Bones
Dorcas Meadowes
Doris Crockford
Elphias Doge
Ted
Andromeda
Bellatrix
Rodolphus
Narcissa
Draco
Lucius
Sirius
Regulus
Kreacher
Walburga
Orion
Alphard
Dorea
Charlus
Fleamont
Euphemia
James
Lily
Petunia
Vernon
Dudley
Harry
Albus
Aberforth
Ariana
Kendra
Percival
Muriel
Elphinstone
Rowena
Helga
Godric
Salazar
Nicolas
Perenelle
Minerva
Quirinus
Quirrell
Severus
Lily
James
Harry
Voldemort
Dumbledore
Albus
Aberforth
Ariana
Kendra
Percival
Muriel
Elphinstone
Rowena
Helga
Godric
Salazar
Nicolas
Perenelle
Minerva
Quirinus
Quirrell
Severus
Lily
James
Harry
Voldemort
Dumbledore

GLOBAL


Rocklie Portner
Agatha Trams
Cassandra Trelawney
Montblanc
Otto
Patricia
Ragnok
Marcus
Graham
Neville
Kenneth
Montgomery
Apolline
Otto
Frank Longbottom
Millicent Bagnold
Edgar Bones
Cassandra Vablatsky
Tatting
Augusta
Twilfitt
Sir Nicholas de Mimsy-Porpington
Ludo Bagman
Mr. Ollivander
Mr. Gringott
Mr. Dursley
Mrs. Norris
Mrs. Weasley
Mrs. Granger
Mrs. Dursley
Mrs. Bulstrode
Mrs. Malkin
Mrs. Cole
Mrs. Figg
Mrs. Cattermole
Mrs. Skower
Mrs. Hudson
Mrs. Cake
Mrs. Puddifoot
Mrs. Cadwallader
Mrs. Scamander
Mrs. Longbottom
Mrs. Flume
Mrs. Crabbe
Mrs. Goyle
Mrs. Lovegood
Mrs. Nettles
Mrs. Bell
Mrs. Hall

Network analysis and results

Medium density

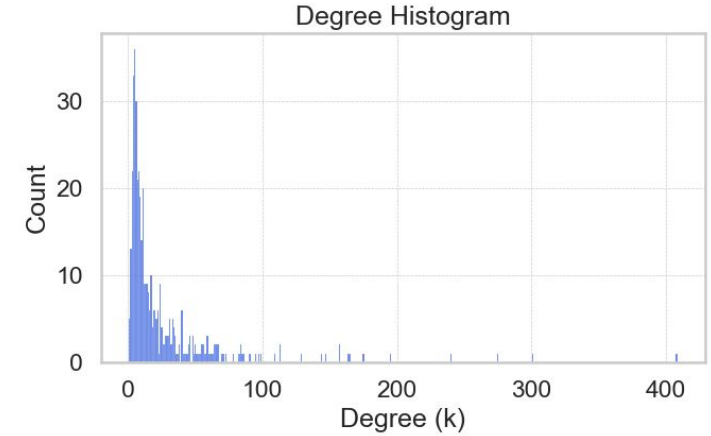
High average clustering

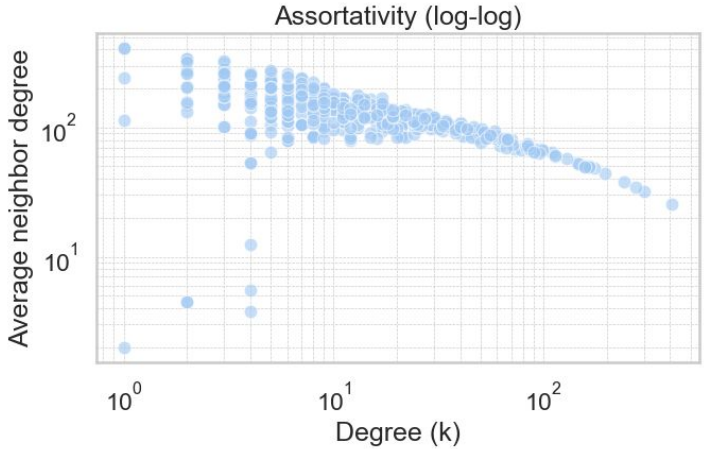
N	L	$\mathbf{d}$	$\langle k \rangle$	$k_{\mathbf{max}}$	$k_{\mathbf{min}}$	$\langle C \rangle$	$\langle d \rangle$	d_{max}	r
440	5444	0.056	24.75	408	1	0.79	2.02	4	-0.33

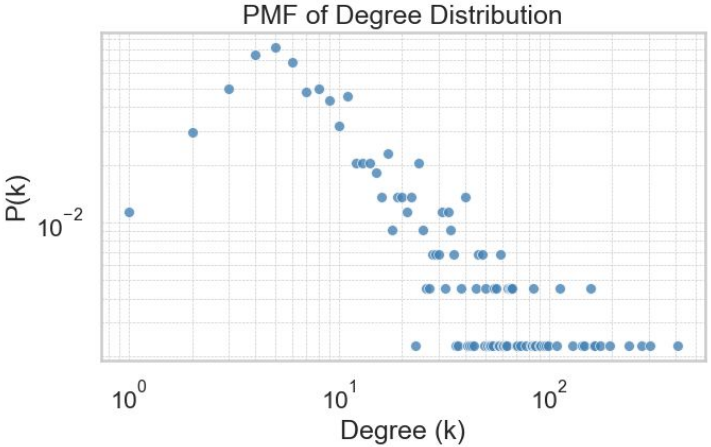
Close to N

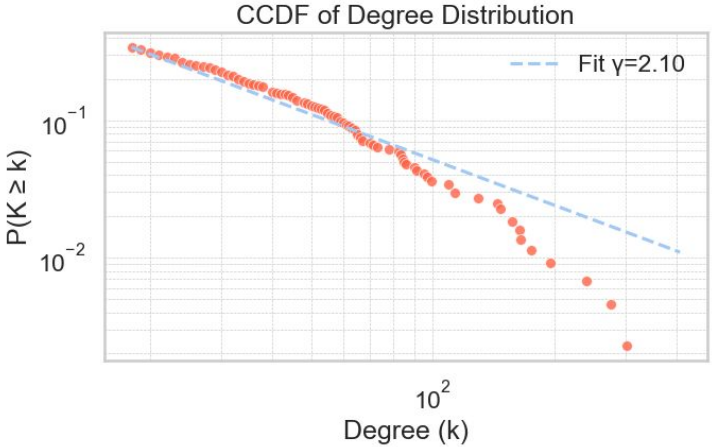
Dissasortative

Degree Distribution of the Network









node	degree	local_clustering	degree_centrality	betweenness_centrality	closeness_centrality
Harry	322	0.058706	0.759434	0.408071	0.803030
Harvey	1	0.000000	0.002358	0.000000	0.445846
Dumbledore	192	0.125164	0.452830	0.084057	0.634731
Voldemort	111	0.233743	0.261792	0.018154	0.561589
Godric	26	0.523077	0.061321	0.005172	0.480726
			...		
Walden	11	1.000000	0.025943	0.000000	0.465934
Molly Weasley	4	1.000000	0.009434	0.000000	0.448203
Scorpius	1	0.000000	0.002358	0.000000	0.397749
Victoire	2	1.000000	0.004717	0.000000	0.320242
Hugo	1	0.000000	0.002358	0.000000	0.445846

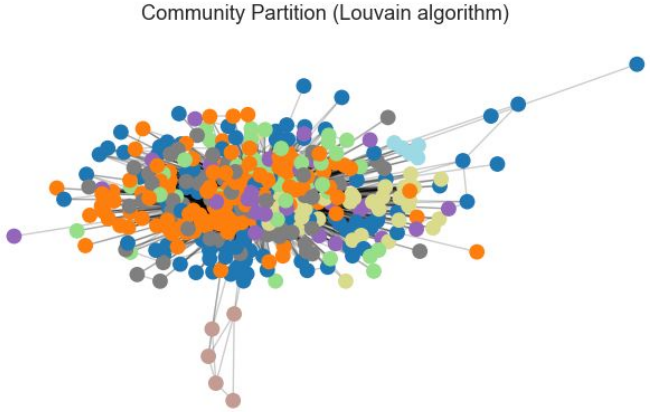
Table 4.3: Node-level metrics (degree, clustering, and centralities) for the global character network. Only the first and last five nodes are shown.

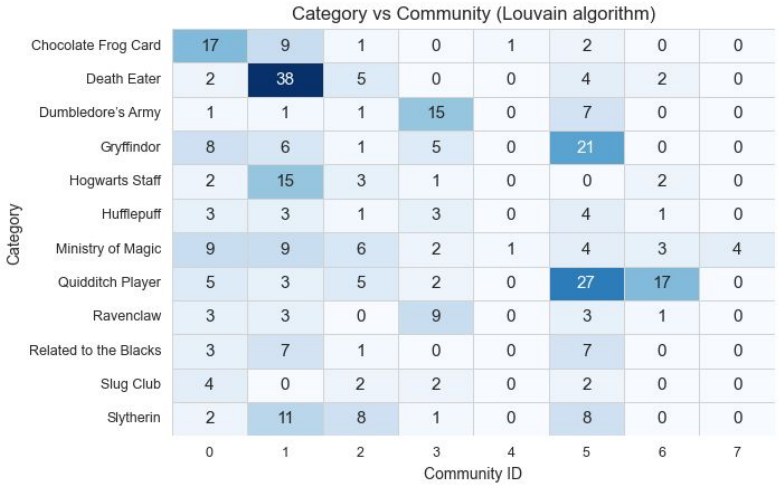
Communities

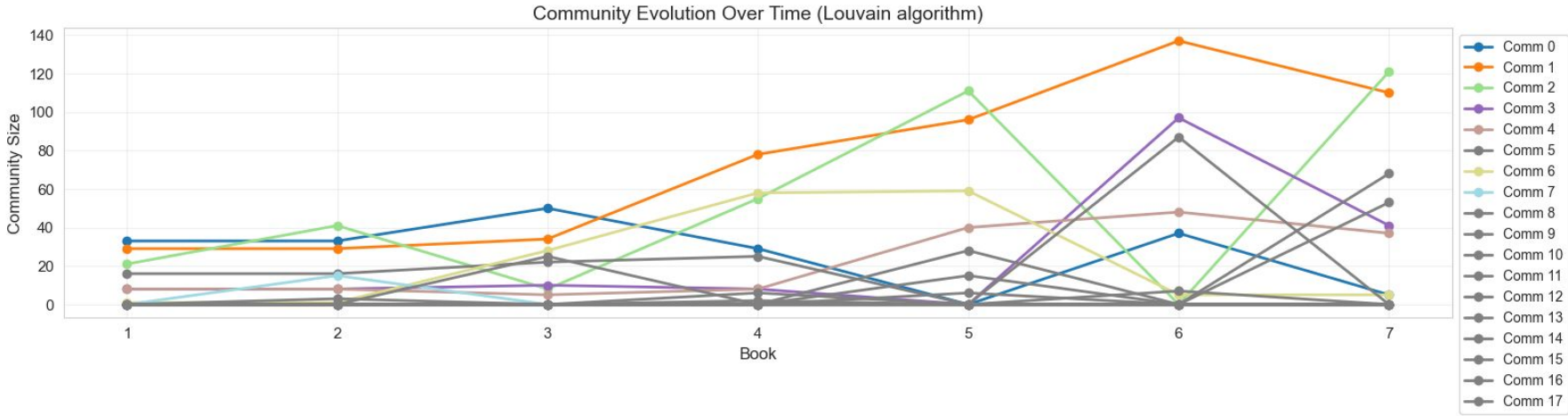
Non-overlapping communities (Louvain method)

Community ID	Size	Internal Edges	Density	Avg. Internal Degree	Avg. External Degree
1	121	785	0.108	12.975	12.736
0	110	376	0.063	6.836	10.727
5	68	352	0.155	10.353	20.309
2	53	183	0.133	6.906	19.717
3	41	194	0.237	9.463	25.366
6	37	161	0.242	8.703	14.757
4	5	7	0.700	2.800	1.000
7	5	10	1.000	4.000	2.800

Table 4.3: Summary of Louvain communities detected in the global character network.

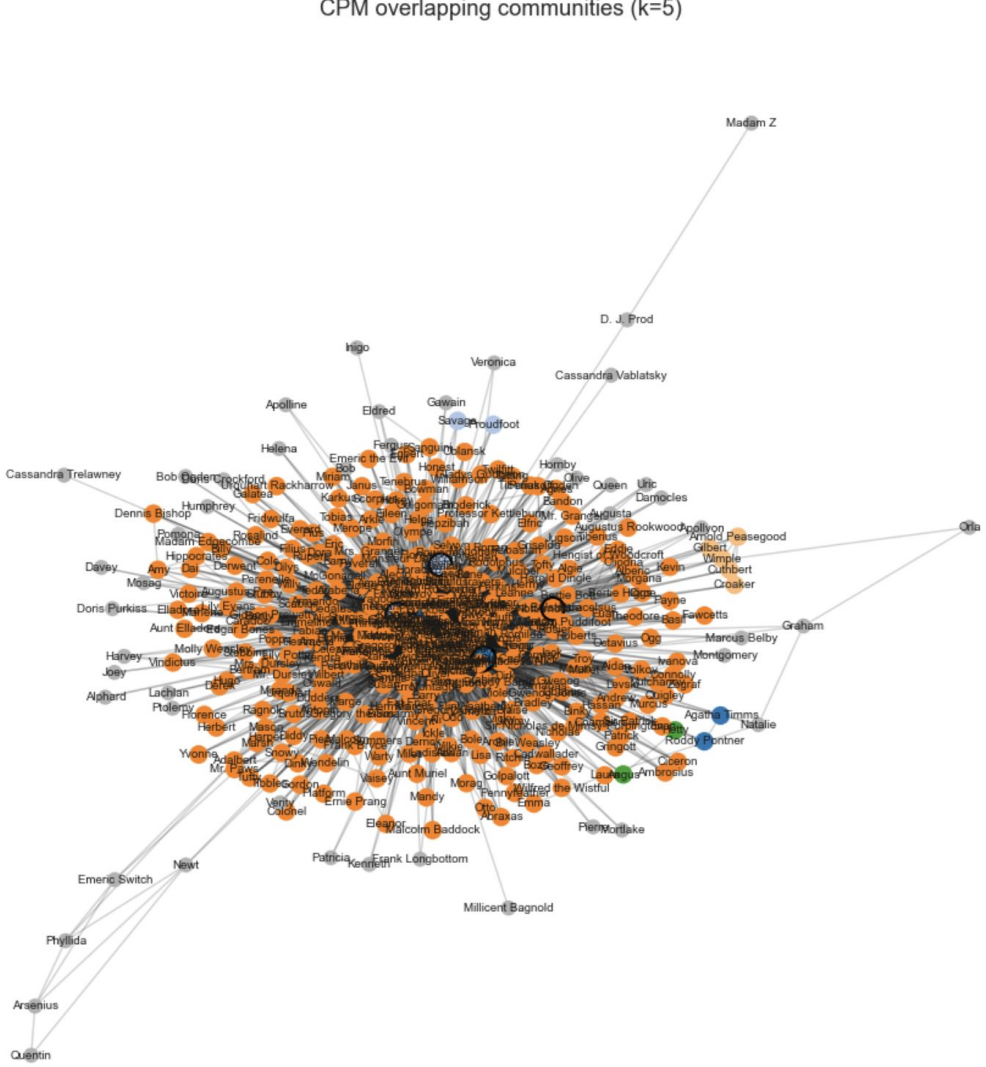

Community Partition (Louvain algorithm)


Category vs Community (Louvain algorithm)
Chocolate Frog Card 17 9 1 0 1 2 0 0
Death Eater 2 38 5 0 0 4 2 0
Dumbledore's Army 1 1 1 15 0 7 0 0
Gryffindor 8 6 1 5 0 21 0 0
Hogwarts Staff 2 15 3 1 0 0 2 0
Hufflepuff 3 3 1 3 0 4 1 0
Ministry of Magic 9 9 6 2 1 4 3 4
Quidditch Player 5 3 5 2 0 27 17 0
Ravenclaw 3 3 0 9 0 3 1 0
Related to the Blacks 3 7 1 0 0 7 0 0
Slug Club 4 0 2 2 0 2 0 0
Slytherin 2 11 8 1 0 8 0 0
Category
0 1 2 3 4 5 6 7
Community ID

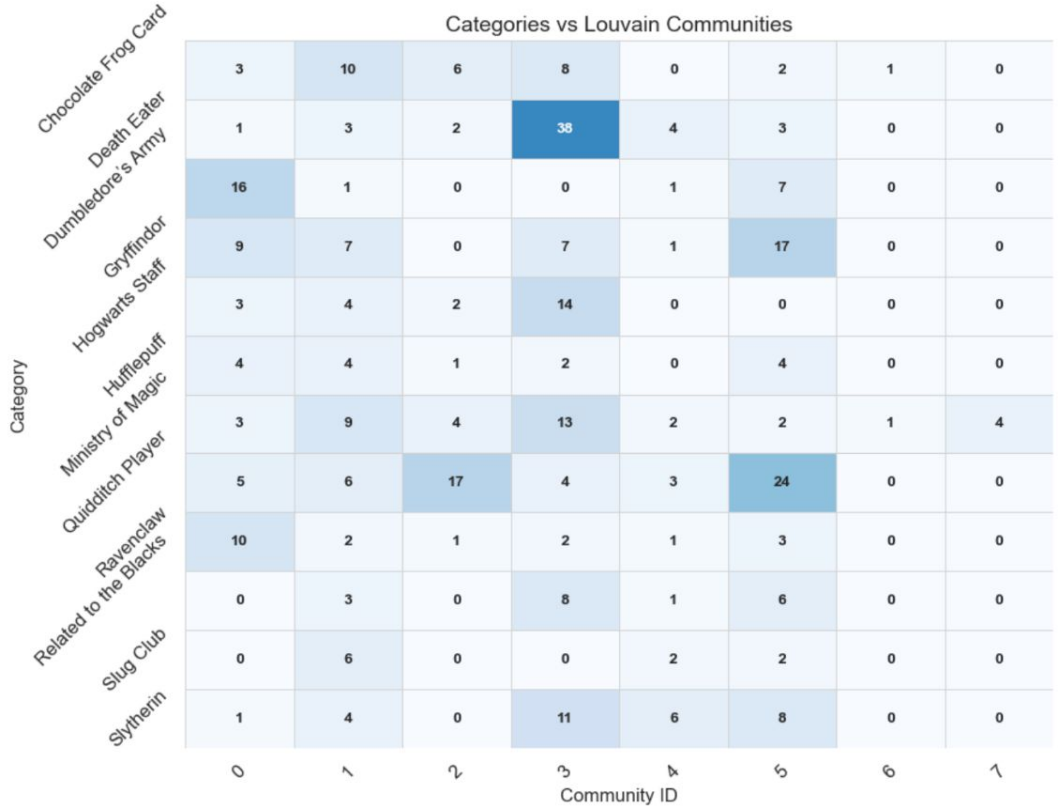

Community Evolution Over Time (Louvain algorithm)
Community Size
140 120 100 80 60 40 20 0
1 2 3 4 5 6 7
Book
Comm 0
Comm 1
Comm 2
Comm 3
Comm 4
Comm 5
Comm 6
Comm 7
Comm 8
Comm 9
Comm 10
Comm 11
Comm 12
Comm 13
Comm 14
Comm 15
Comm 16
Comm 17

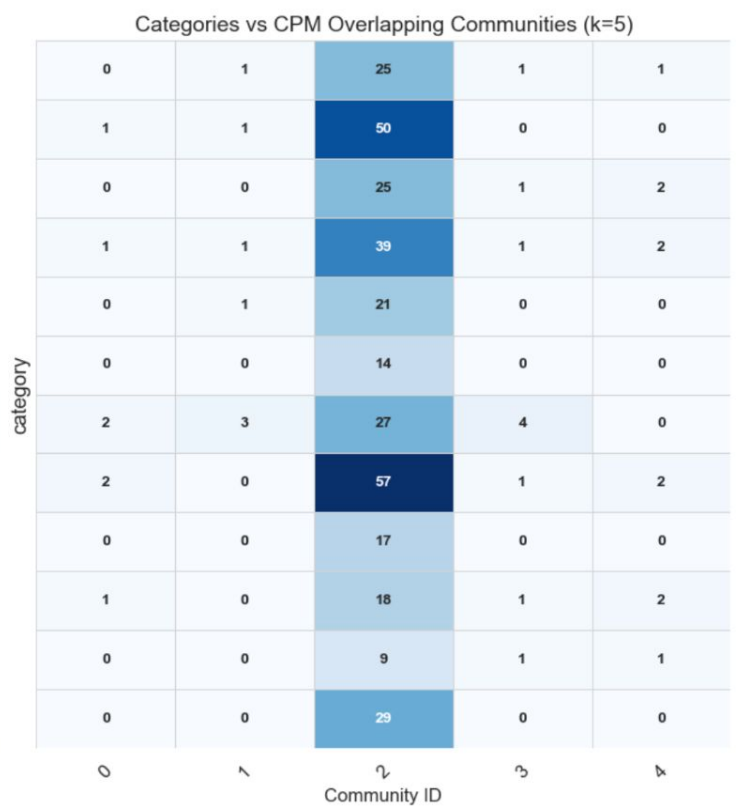
Community ID	Size	Num. Cliques	Density	Avg. Degree	Overlapping Nodes
0	5	1	1.000000	4.000000	3
1	5	1	1.000000	4.000000	3
2	381	2800	0.072952	27.721785	11
3	8	2	0.964286	6.750000	3
4	5	1	1.000000	4.000000	3

Table 4.4: Extended CPM community statistics for $k = 5$ in the global network. The table reports community sizes, number of 5-cliques, subgraph density, average internal degree, and number of nodes shared across multiple CPM communities.


CPM overlapping communities (k=5)
Madam Z
D. J. Prod
Inigo
Veronica
Cassandra Vablatsky
Apolline
Eldred
Gawain
Savage
Proudfoot
Halena
Cassandra Trelawney
Bob
Crockford
Rackharrow
Janus
Tenebrus
Honest
Homby
Olive
Queen
Uric
Damocles
Dennis Bishop
Humphrey
Fridwulfa
Tobias
Meropey
Helpa
Professor
Kettleburn
Elfric
Mr. Granger
Augusta
Augustus Rookwood
Apollyon
Arnold Peasegood
Gilbert
Wimple
Orla
Davey
Amy
Dai
Derwent
Lilly
Hippocrates
Madam Edgecombe
Rosalind
Albert
Kevin
Cuthbert
Morgana
Croaker
Mosag
Victoire
Augustus
Doris Purkiss
Elladora
Marsha
Aunt Elladora
Edgar
Bones
Payne
Fawcetts
Theodore
Graham
Marcus Belby
Octavius
Ogg
Montgomery
Harvey
Molly Weasley
Vindictus
Joey
Durslev
Wilbert
Ivanova
Alphard
Lachlan
Dolohov
Florence
Ragnok
Gruttock
Agatha Timms
Natalie
Roddy Pontner
Patrick
Gringott
Yvonne
Adalbert
Mr. Payne
Gordon
Platform
Wendelin
Snowy
Dink
Yaxley
Aunt Muriel
Ambrosius
Cicero
Laurence
Wilfred the Wistful
Emma
Ernie Prang
Mandy
Colonel
Abraxas
Eleanor
Malcolm Baddock
Mortlake
Patricia
Kenrelin
Frank Longbottom
Newt
Emeric Switch
Millicent Bagnold
Phyllida
Arsenius
Quentin

Category exploration by community

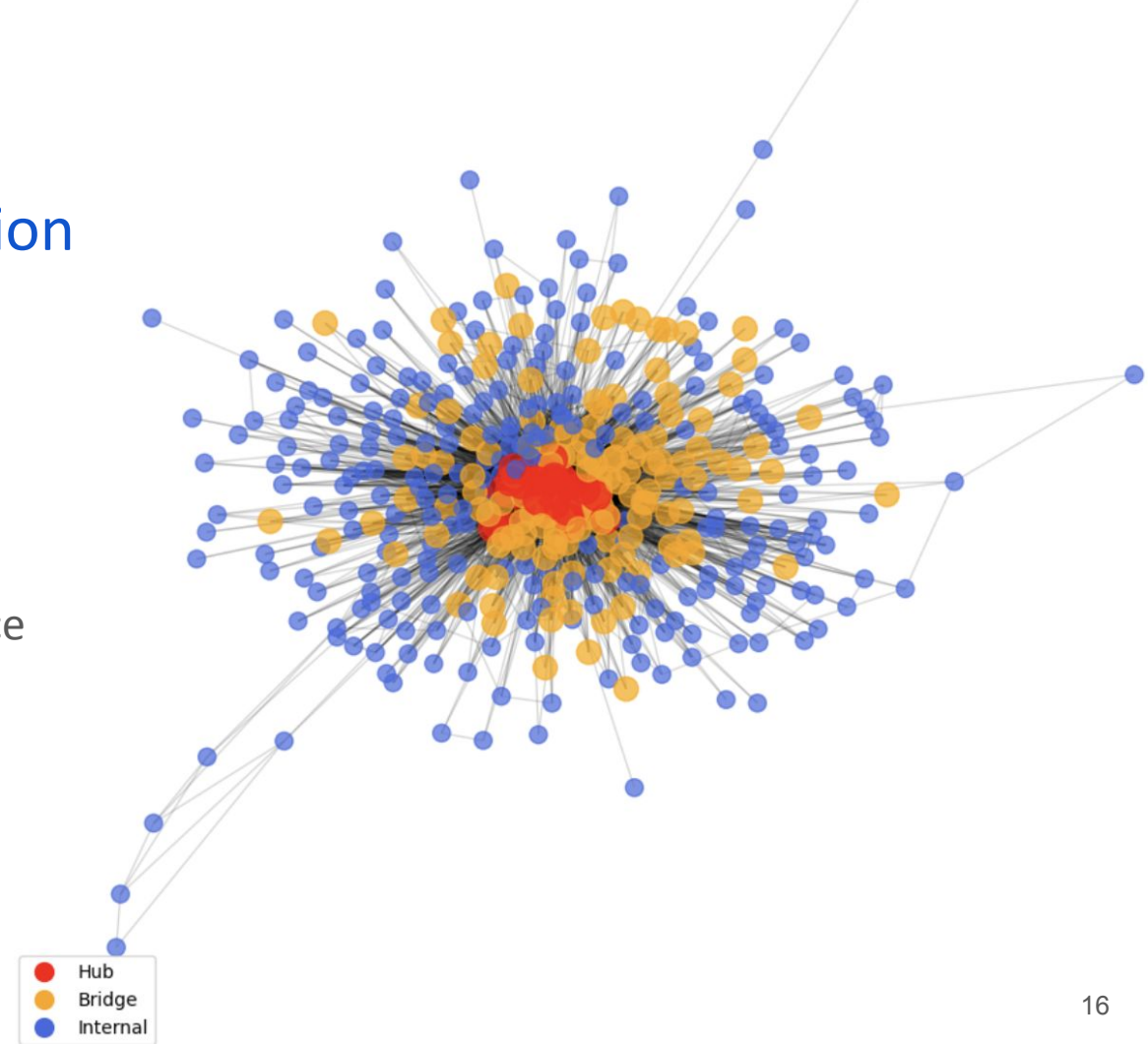




Phenomenon exploration

HUBS

- Great narrative importance
- Spread of information
- Plot development

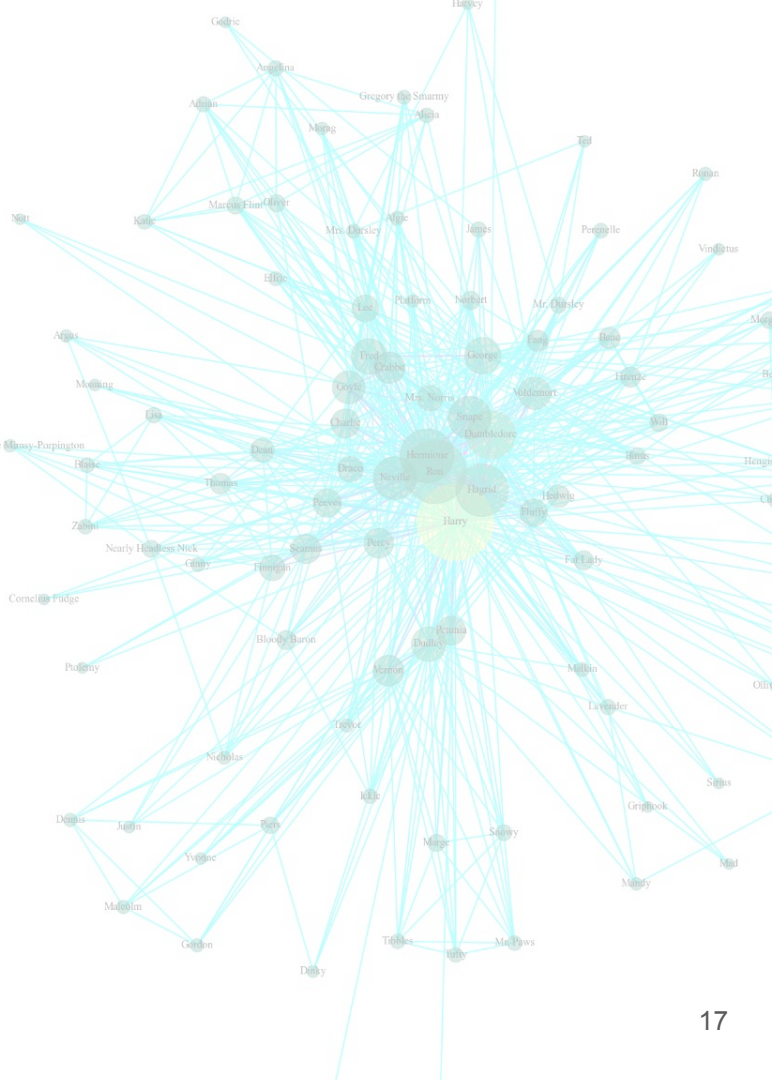


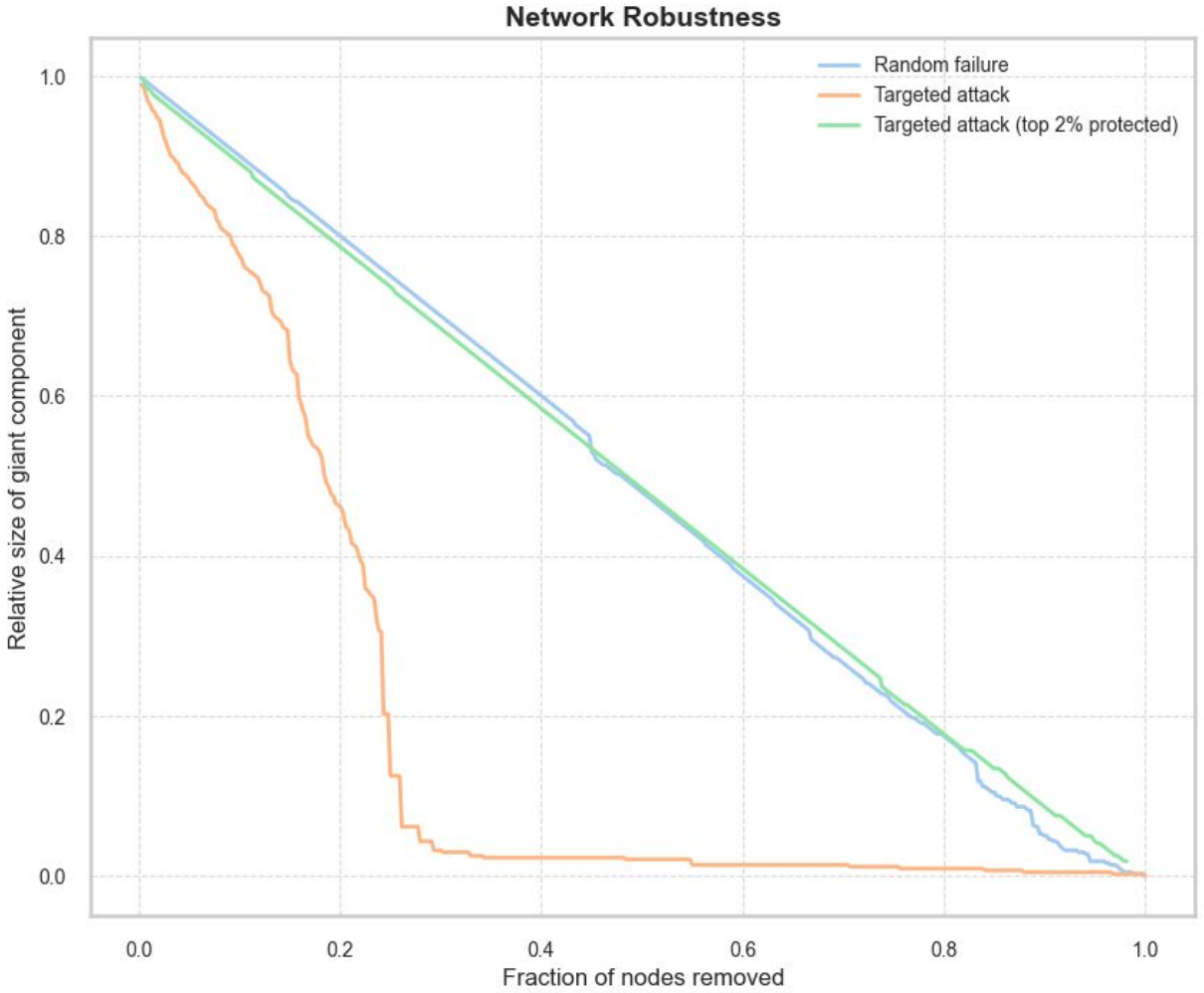
Small world property

Tightly interconnected structure

$$\langle d \rangle = 2.02$$

$$\frac{\ln N}{\ln \langle k \rangle} = \frac{\ln(440)}{\ln(24.745)} \approx \frac{6.086}{3.21} \approx 1.89.$$

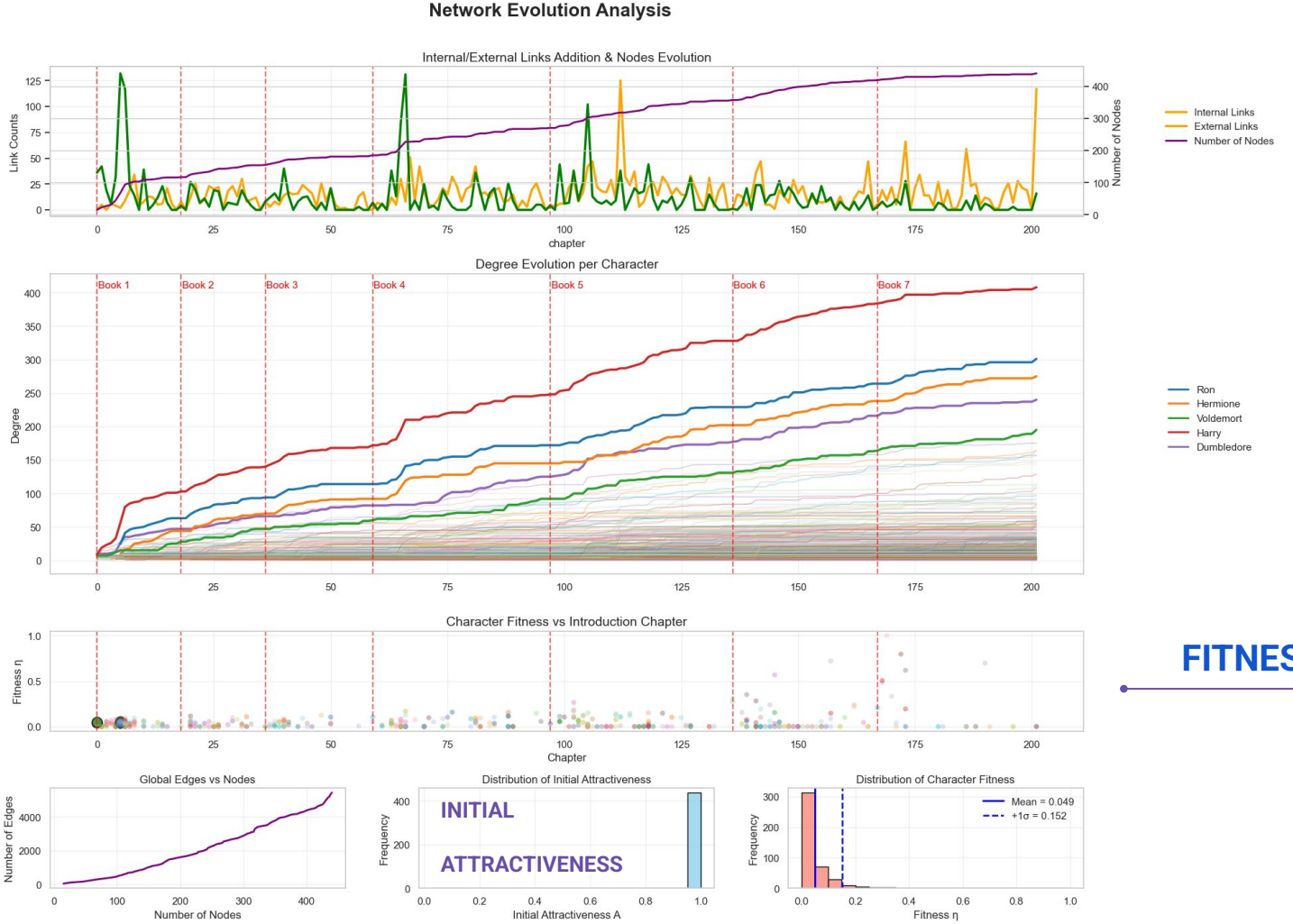



Network Robustness
Random failure
Targeted attack
Targeted attack (top 2% protected)
Relative size of giant component
Fraction of nodes removed
0.0
0.2
0.4
0.6
0.8
1.0

INTERNAL LINKS

$\gamma \approx 2$

ACCELERATED GROWTH



FITNESS

Model comparisons and interpretation

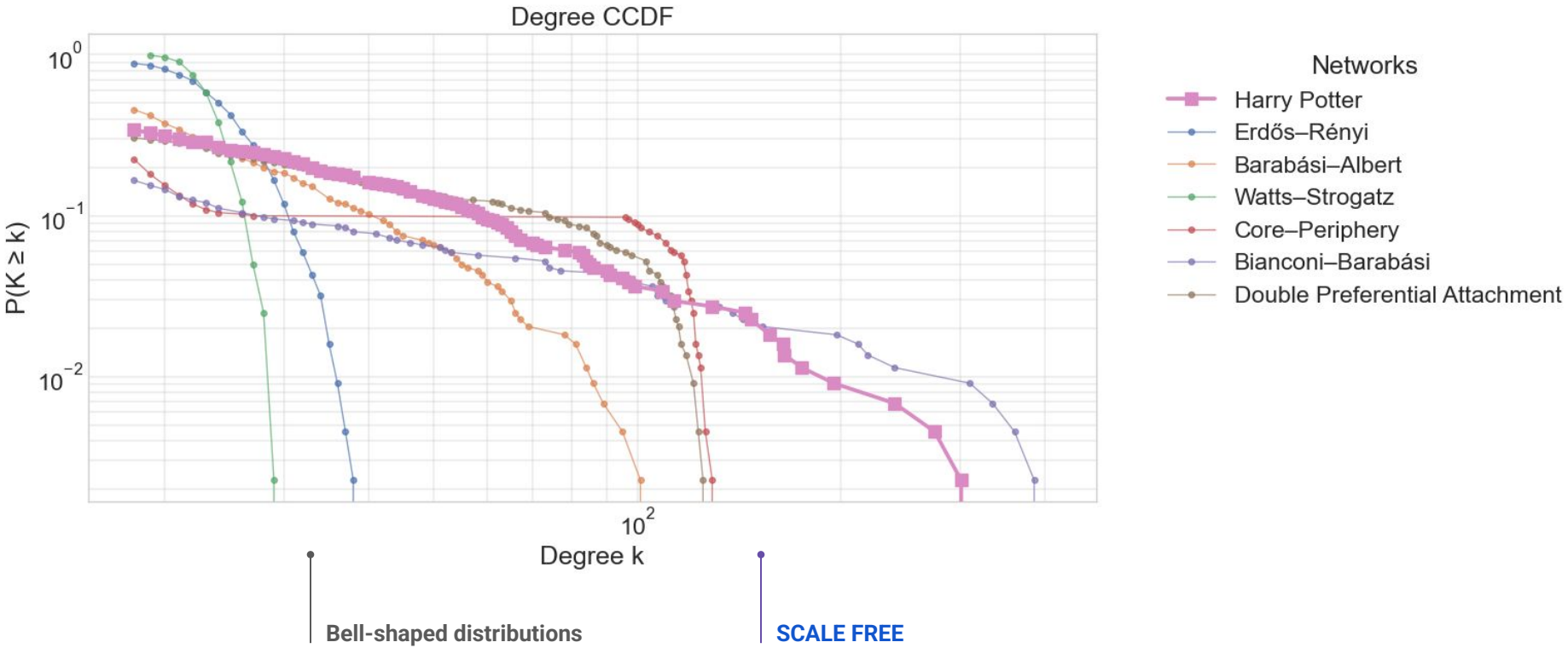
All model parameters estimated using the Harry Potter's network

EXAMPLE (BA)

$$m = \lfloor E/N \rfloor = 12, \quad N_0 = \lfloor N \cdot \text{main_char_percent} \rfloor = 22$$

Network	L	$\langle k \rangle$	$k_{\mathbf{max}}$	$k_{\mathbf{min}}$	$\langle C \rangle$	$\langle d \rangle$	r	Connected	$\hat{\gamma}$
Harry Potter	5444	24.75	408	1	0.79	2.02	-0.33	True	2.10
Erdős–Rényi	5440	24.73	39	9	0.05	2.18	0.00	True	–
Watts–Strogatz	5280	24.00	30	19	0.37	2.48	0.01	True	–
Core–Periphery	5403	24.56	131	6	0.38	2.07	-0.29	True	–
Barabási–Albert	5037	22.90	106	8	0.10	2.22	-0.04	True	3.00
Bianconi–Barabási	5202	23.65	406	12	0.67	1.96	-0.44	True	2.11
D. P. Attachment	5189	23.59	138	5	0.37	2.28	-0.12	True	2.02

Table 6.1: Network statistics for generated models and the Harry Potter graph.


Degree CCDF
10^0
10^{-1}
10^{-2}
P(K ≥ k)
10^2
Degree k
Networks
Harry Potter
Erdős–Rényi
Barabási–Albert
Watts–Strogatz
Core–Periphery
Bianconi–Barabási
Double Preferential Attachment
Bell-shaped distributions
SCALE FREE

Conclusions and further work

- Best model: **Bianconi-Barabási**
- Network science as a tool in storytelling and screenwriting
- Enhanced comprehension and narrative design
- Improvements
  - Graph generation method (LLMS, name entity recognition)
  - Directed networks and sentiment analysis to label relationships
- Extend to other series to explore a general narrative structure
- Comparative analysis to create writing aid

References

[1] Albert-László Barabási. Network Science. Cambridge, UK: Cambridge University Press, 2016.

[2] Stephen P. Borgatti and Martin G. Everett. “Models of core/periphery structures”. In: Social Networks
21.4 (2000), pp. 375–395.

[3] Full Character Listing — Wizardry / Magical-Menagerie.com. url: https : / / magical - menagerie .
com/wizardry/full-character-listing/ (visited on 11/30/2025).

[4] Shubham Maindola. Harry Potter Books Dataset. Kaggle. [Data set]. url: https://www.kaggle.com/
datasets/shubhammaindola/harry-potter-books (visited on 11/30/2025).

[5] M. E. J. Newman. Networks: An Introduction. Oxford University Press, 2010.